

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/341932912>

Occurrence, Antimicrobial Resistance Patterns, and Genetic Characterization of *Staphylococcus aureus* Isolated from Raw Milk in the Dairy Farms over Two Seasons in China

Article in *Microbial drug resistance* (Larchmont, N.Y.) · June 2020

DOI: 10.1089/mdr.2019.0358

CITATIONS

13

READS

203

9 authors, including:



Harvey Ho

University of Auckland

105 PUBLICATIONS 1,085 CITATIONS

SEE PROFILE



S. M. E. Rahman

Bangladesh Agricultural University

100 PUBLICATIONS 2,486 CITATIONS

SEE PROFILE



Rongwei Han

Qingdao Agricultural University

100 PUBLICATIONS 1,932 CITATIONS

SEE PROFILE

Occurrence, Antimicrobial Resistance Patterns, and Genetic Characterization of *Staphylococcus aureus* Isolated from Raw Milk in the Dairy Farms over Two Seasons in China

Cuiping Shi,^{1,†} Zhongna Yu,^{2,†} Harvey Ho,³ Jun Wang,¹ Wei Wu,¹ Mengru Xing,¹ Yutao Wang,⁴ S.M.E. Rahman,⁵ and Rongwei Han¹

This study investigated the occurrence and resistance rates of *Staphylococcus aureus* strains isolated from raw milk in the dairy farms over two seasons (spring and autumn) and across four regions that included 11 provinces in China. In total, 750 raw milk samples from the 405 dairy farms were collected. Fifteen antimicrobial agents were tested for antimicrobial resistance via disk diffusion tests, and PCR tests were performed to identify drug resistance genes of *S. aureus* isolates. Out of 750 samples, 276 (36.8%) were positive for *S. aureus*, with 150 (41.1%) being positive in spring and 126 (32.7%) being positive in autumn. The occurrence rate of *S. aureus* in northeastern China (45%) was higher than that in western China (33%) and southern China (31.9%), respectively, and the rate significantly ($p < 0.05$) differed from those of western China and southern China. Of 276 isolates, 261 (94.6%) strains were resistant to more than 1 antimicrobial drug, and 193 (69.9%) strains were multidrug resistant. The *blaZ* (46.3%), *dfrG* (35.5%), and *tetM* (27.2%) genes were detected at a high frequency in the *S. aureus* strains. Our data revealed a variation ($p < 0.05$) in the resistance patterns in the different regions and across the two seasons. The occurrence and drug resistance rates of *S. aureus* isolated from raw milk obtained from dairy farms may still cause severe problems in China.

Keywords: raw milk, dairy farms, *Staphylococcus aureus*, antimicrobial resistance, antimicrobial resistance genes

Introduction

MASTITIS IN DAIRY HERDS is a prevalent global disease that causes economic burdens due to reduced milk production, increased treatment costs, and potential concerns relating to public health.^{1–4} A recent study has also indicated that for bovine mastitis cases, a few species of bacteria, such as *Staphylococcus aureus*, *Streptococcus agalactiae*, *Streptococcus dysgalactiae*, *Streptococcus uberis*, coagulase-negative staphylococci, and *Escherichia coli*, predominate.⁵ Of several etiological agents, *S. aureus* is the most common mastitis-causing pathogen in dairy cows.^{6,7} Due to its ubiquity, *S. aureus* exists in food-processing environments and causes foodborne poisoning and infections in both humans

and animals.^{8,9} Ultimately, the risk of milk contamination can increase after dairy products reach the consumer market.¹⁰

Antimicrobial treatment is one recognized method used to prevent and treat dairy cow mastitis.^{11,12} However, the use of antimicrobials for infection treatment may lead to the potential development of antimicrobial resistance.¹³ A previous study reported that 30% of *S. aureus* strains isolated from raw mastitic milk samples were resistant to one or more classes of antimicrobial agents.¹⁴ In a similar manner, cows in China also suffer from mastitis caused by resistant *S. aureus*. For example, among the 28.0% of *S. aureus* strains isolated from bovine mastitic milk in Zhejiang province, 90.7% of 75 representative *S. aureus* isolates were resistant to at least one antimicrobial class.¹⁵

¹College of Food Science and Engineering, Qingdao Agricultural University, Qingdao City, China.

²Haidu College, Qingdao Agricultural University, Laiyang, China.

³Auckland Bioengineering Institute, The University of Auckland, Auckland, New Zealand.

⁴Institute of Quality Standard and Test Technology for Agro-products, Shandong Academy of Agricultural Sciences, Jinan, China.

⁵Department of Animal Science, Bangladesh Agricultural University, Mymensingh, Bangladesh.

[†]These authors contributed equally.

The *staphylococci* that originate in animals usually harbor a wide variety of antimicrobial resistance genes.¹⁶ In recent years, many reports have confirmed that *S. aureus* isolated from the milk of cows diagnosed with mastitis carries multiple drug resistance genes. A report from the United States indicated that 27.7% of *S. aureus* isolates carried multiple resistance genes (mainly *blaZ* combined with *tet*).¹² In Iran, *blaZ* (46%) and *tetM* (34.8%) resistance genes were found at a high frequency in *S. aureus* isolated from raw milk (from cows and sheep) and dairy products.¹⁷ In northwest China, genotyping revealed that the most prevalent resistance genes carried by *S. aureus* isolated from bovine mastitic samples were *rpoB* (100%) and *blaZ* (95.45%).¹⁸ Therefore, monitoring of the antimicrobial characterization of mastitis-causing *S. aureus* isolates is required for the antibiotic treatment of animals on dairy farms.¹⁹

To the best of our knowledge, studies of the occurrence and characterization of *S. aureus* cultured from raw milk in the dairy farms have mainly focused on individual areas of China; however, a little information currently exists about the regional and seasonal changes of *S. aureus* isolated from raw milk samples in the dairy farms.^{19,20} Thus, the objective of this study was to investigate the occurrence, drug-resistant phenotypes, and genotypes of *S. aureus* isolated from 750 raw milk samples in the 405 dairy farms collected from four regions over two seasons in China. These results may provide clues that can be used to select the best drugs and treatments for dairy farm cows with mastitis.

Materials and Methods

Collection of samples

In total, 750 raw milk samples (with a somatic cell count >200,000 cells/mL in the milk) were collected from 405 major dairy farms in 11 provinces of China, including northeastern China ($n=200$, Heilongjiang and the Inner Mongolia provinces), northern China ($n=240$, Hebei, Beijing, Tianjin, and Shandong), western China ($n=100$, Xinjiang), and southern China ($n=210$, Shanghai, Jiangxi, Anhui, and Sichuan provinces). Dairy farms were selected if the farm owned over 200 dairy cows and also based on where the cows were housed, grouped, and managed according to their ages and lactation cycle and if the cows were fed with total mixed rations. In addition, sand or organic bedding had to be used for the herds. The samples were collected from the same dairy farms over two seasons: spring ($n=365$, March–April 2016) and autumn ($n=385$, September–October 2016). One milk sample consisted of a mix of milk from 8 to 10 individual dairy cows with a somatic cell count >200,000 cells/mL in the milk from each farm. These samples were collected and then used for the investigation.

A 50-mL milk sample was taken and poured into a sterile tube after the teats of the cow had been disinfected with 0.5% iodine solution (Merck, Germany) and wiped dry with a clean paper towel. The first four to five streams of the foremilk sample were removed before the actual sample was collected. All milk samples were stored at 4°C and transported to the laboratory within 24 hr for the experiment. The experimental procedures in the present study were approved by the Animal Care Advisory Committee at Qingdao Agricultural University.

Isolation and identification of *S. aureus*

The procedure for the *S. aureus* isolation and identification was carried out based on a previous method.¹⁷ In brief, 10 mL of each raw milk sample was added to a 90-mL trypticase soy broth (Qingdao Haibo Biotechnology Co. Ltd., Qingdao, China). The solution was incubated at 37°C in an incubator for 24 hr. After preenrichment, the culture was inoculated on Baird-Parker agar supplemented with 5% egg yolk and tellurite (Baird-Parker agar; Beijing Land Bridge Technology Ltd., Beijing, China) at 37°C for 24–48 hr. The colonies with a typical black appearance surrounded by a clear zone were considered to be *S. aureus*. The *nuc* gene-based PCR method was used to confirm the presence of *S. aureus*.⁹ The confirmed *S. aureus* strains were preserved at –80°C for further study.

Antimicrobial susceptibility test

The *S. aureus* isolates were subjected to antimicrobial susceptibility testing using the Kirby-Bauer disk diffusion method on Mueller–Hinton Agar (Qingdao Haibo Biotechnology Co. Ltd.) according to the guidelines of the Clinical Laboratory Standards Institute.²¹ Fifteen total commercial antibiotic disks (Tianhe Microbial Reagent Co. Ltd., Hangzhou, China) were used in the experiment and included disks with penicillin G (10 U), cefoxitin (30 µg), oxacillin (1 µg), erythromycin (15 µg), ciprofloxacin (5 µg), doxycycline (30 µg), tetracycline (30 µg), trimethoprim (5 µg), sulfamethoxazole (300 µg), gentamicin (10 µg), tobramycin (10 µg), kanamycin (30 µg), amikacin (30 µg), clindamycin (2 µg), and chloramphenicol (30 µg). *S. aureus* ATCC 25923 was used as the positive control.

Detection of antimicrobial resistance genes

The DNA of each *S. aureus* isolate was obtained using the boiling method. First, the live or dead *S. aureus* isolate cells were collected by centrifugation at 10,000 *g* for 5 min. Then, the supernatant was discarded, and 200 µL of 1× phosphate-buffered saline-Tween 20 buffer was added to the tube, vortexed, and mixed. The tube was boiled in water for 10 min and then quickly cooled in an ice bath for 5 min. Finally, the supernatant was retained as a DNA template after centrifugation at 10,000 *g* for 5 min. The DNA templates were stored at –80°C before testing.

The antimicrobial resistance genes of all the isolates were amplified by PCR (BIO-RAD MJ Mini PCR System, CA). Table 1 shows all the PCR primer information. The primers were synthesized by Sangon Biotech Co., Ltd. (Shanghai, China) and included penicillin G (*blaZ*), cefoxitin (*cfxA*), oxacillin (*mecA*), erythromycin (*ermA*, *ermB*, *ermC*, *msrA*, *msrB*, and *mphC*), tetracycline (*tetL*, *tetM*, and *tetK*), trimethoprim (*dfrG*, *dfrK*, and *dfrS1*), aminoglycosides (*aac(6′)-aph(2′)*), *ant(6)-Ia*, and *ant(4′)-Ia*), lincosamides (*lnuA* and *lnuB*), chloramphenicol (*cat::pC194*, *cat::pC221*, *cat::pC223*, and *fexA*) resistance genes, and the multidrug resistance gene (*cfr*). The PCR mixture (25 µL) consisted of 0.3 µL of Taq DNA polymerase, 0.4 µL of each primer, 1 µL of the DNA template, 2.2 µL of 2.5 mmol/L dNTP, 2.5 µL of 10× the PCR buffer, and 18.2 µL of ddH₂O. All PCR amplified products were analyzed by 0.8% (w/v) agarose gel electrophoresis and stained with 4S Red Plus nucleic acid

TABLE 1. PRIMERS USED FOR ANTIMICROBIAL RESISTANCE GENES DETECTION IN THIS STUDY

<i>Antibiotic classes</i>	<i>Gene</i>	<i>Primers</i>	<i>DNA sequence 5' → 3'</i>	<i>Amplified product (bp)</i>	<i>References</i>
β-Lactams	<i>nuc</i>	nuc-F	GCGATTGATGGTGATACGGTT	279	22
		nuc-R	AGCCAAGCCTTGACGAACATAAAGC		
	<i>blaZ</i>	blaZ-F	ACTTCAACACCTGCTGCTTTC	173	22
		blaZ-R	TGACCACTTTTATCAGCAACC		
	<i>cfxA</i>	cfxA-F	GCTCAACACAGATAGTTTAT	312	23
Macrolides		cfxA-R	GAGTCACAATGATGTGCC		
	<i>mecA</i>	mecA-F	AAATCGATGGTAAAGGTTGGC	532	24
		mecA-R	AGTTCTGCAGTACCGGATTGC		
	<i>ermA</i>	ermA-F	AAGGGTAAACCCCTCTGA	190	24
		ermA-R	TTCGCAATCCCTTCTCAAC		
	<i>ermB</i>	ermB-F	CTATCTGATTGTTGAAGAAGGATT	142	25
		ermB-R	GTTTACTCTTGGTTTAGGATGAAA		
	<i>ermC</i>	ermC-F	CTTGTGATCAGCATATTTCC	190	25
		ermC-R	ATCTTTAGCAAAACCCGTATTC		
	<i>msrA</i>	msrA-F	TCCAATCATTGCACAAAATC	163	25
Tetracyclines		msrA-R	AATCCCTCTATTGGTGGT		
	<i>msrB</i>	msrB-F	TATGATATCCATAATAATATCCAATC	595	26
		msrB-R	AAGTATATCATGAATAGATTGTCTGTT		
	<i>mphC</i>	mphC-F	GAGACTACCAAGAAACCTGACG	722	27
		mphC-R	CATACGCCGATTCTCCTGAT		
	<i>tetL</i>	tetL-F	CATTGGTCTTATTGGATCG	475	19
		tetL-R	ATTACAGTTCCGATTTCGG		
Sulfonamides	<i>tetM</i>	tetM-F	AGTGGAGCGATTACAGAA	158	22
		tetM-R	CATATGTCTCTGGCGTGTCTA		
	<i>tetK</i>	tetK-F	GTAGCGACAATAGGTAATAGT	360	22
		tetK-R	GTAGTGACAATAAACCTCCTA		
	<i>dfrG</i>	dfrG-F	TGCTGCGATGGATAAGAA	405	26
		dfrG-R	TGGGCAATACCTCATTC		
	<i>dfrK</i>	dfrK-F	CAAGAGATAAGGGTTCACG	229	26
Aminoglycosides		dfrK-R	ACAGATACCTTCGTTCCACTC		
	<i>dfrS1</i>	dfrS1-F	CACCTGTAATGGCACGGAAA	270	26
		dfrS1-R	CGAATGIGTATGGTGGAAAG		
	<i>aac(6')-aph(2'')</i>	aac(6')-aph(2'')-F	CAGAGCCTTGGGAAGATGAAG	348	19
		aac(6')-aph(2'')-R	CCTCGTGAATTCAATGTTCTGGC		
	<i>ant(6)-Ia</i>	ant(6)-Ia-F	ACTGGCTTAATCAATTTGGG	577	19
		ant(6)-Ia-R	GCCTTCCGCCACCTCACCG		
	<i>ant(4')-Ia</i>	ant(4')-Ia-F	CTGCTAAATCGGTAGAAAGC	172	26
		ant(4')-Ia-R	CAGACCAATCAACATGGCACC		
	<i>lnuA</i>	lnuA-F	GGTGGCTGGGGGTAGATGATTAACCTGG	323	28
Lincosamides		lnuA-R	GCTTCTTTTGAATAACATGGTATTTTCGATC		
	<i>lnuB</i>	lnuB-F	CCTACCTATTGTTGTGAA	925	29
		lnuB-R	ATAACGTTACTCTCCTATTC		
	<i>cat::pC194</i>	cat::pC194-F	CAATCCAAGGAATCATTGAAATCGG	472	26
Amphenicols		cat::pC194-R	AAAGCCAGTCATTAGCCTATCTG		
	<i>cat::pC221</i>	cat::pC221-F	TGGAAGTTGTAAATAAAATAAAGTG	269	26
		cat::pC221-R	CAATCCAAGGAATCATTGAAATCGG		
	<i>cat::pC223</i>	cat::pC223-F	AGGATATGAACCTGTATCCTGCTTG	464	26
Multidrug resistance gene		cat::pC223-R	AATAATGAAACATGGTAACCATCAC		
	<i>fexA</i>	fexA-F	GTACTTGTAGGTGCAATACGGCTGA	1272	30
		fexA-R	CGCATCTGAGTAGGACATAGCGTC		
	<i>cfi</i>	cfi-F	TGAAAGTATAAAGCAGGTTGGGAGTCA	746	30
		cfi-R	ACCATATAATTGACCACAAAGCAGC		

dye (BBI Life Science) for 50 min at 80 V and 130 mA, and then the electrophoretic results were analyzed.

Statistical analysis

The data were analyzed using the Chi-square tests available in the Statistical Package for the Social Sciences (SPSS), version 22.0 (SPSS, Inc., Chicago, IL). The regional and seasonal differences among the groups were compared in terms of their occurrence rates, drug resistance rates, and the resistance genes of the *S. aureus* isolates. For all the analyses, a p -value <0.05 was considered statistically significant.

Results

Isolation of *S. aureus* from raw milk samples

In the 750 raw milk samples from the 405 dairy farms, 276 (36.8%) strains of *S. aureus* were isolated and confirmed (Table 2). The occurrence rate of *S. aureus* was significantly higher in raw milk samples from the farms in spring (41.1%) than it was in autumn (32.7%). Ninety *S. aureus* isolates were isolated from the dairy farms in northeastern China, which had the highest isolation rate (45%) among the four regions, followed by 35.8% in northern China and 33% in western China. The lowest isolation rate (31.9%) was detected in southern China. Significant differences in the *S. aureus* isolation rates were found between northeastern and western China ($p < 0.05$), as well as between northeastern and southern China ($p < 0.05$).

Antimicrobial resistance phenotypes of the *S. aureus* isolates

The results of the antimicrobial susceptibility tests demonstrated that 261 (94.6%) *S. aureus* strains from the dairy farms in four regions of China had a severe and variable degree of resistance to 15 tested antimicrobials, including 3 β -lactams, 1 macrolide, 1 quinolone, 2 tetracyclines, 2 sulfonamides, 4 aminoglycosides, 1 lincosamide, and 1 amphenicol (Table 3). The resistance rate of *S. aureus* to five kinds of drugs exceeded 50%, with a resistance rate of 81.5% for penicillin G, 73.6% for sulfamethoxazole, 70.3% for trimethoprim, 63% for oxacillin, and 50.4% for clindamycin. The other drug resistance rates for *S. aureus* were between 14.1% and 37.0%. The most sensitive drug for *S. aureus* was ciprofloxacin, which had the lowest resistance rate of 14.1%. The average *S. aureus* resistance rates to all 15 antimicrobials in northeastern China (40.8%) and northern China (43.5%) were slightly higher than those in western China

(35.1%) and southern China (34.8%), while this rate was slightly higher in spring (43.3%) than it was in autumn (35%). However, no differences in the single-drug resistance rates for *S. aureus* were found among the four regions ($p > 0.05$) or between the two seasons ($p > 0.05$). Regarding the number of drugs with a resistance rate $>50\%$, the four regions were similar. In particular, penicillin G, sulfamethoxazole, and trimethoprim showed resistance in all the different areas. A similar pattern was also found between the two seasons. However, the drug resistance rates of *S. aureus* to penicillin G, oxacillin, sulfamethoxazole, trimethoprim, and erythromycin were significantly higher in spring than in autumn ($p < 0.05$).

Among these isolates, 193 (69.9%) *S. aureus* strains showed multidrug resistance (resistant to at least one agent in at least three antimicrobial classes). One (0.4%) strain was simultaneously resistant to eight classes out of the 14 antimicrobials. The regional and seasonal differences in the multidrug resistance patterns of *S. aureus* are shown in Figs. 1 and 2 and Table 4. The isolates' multidrug resistance patterns in the four regions were similar, with the highest number of strains being resistant to three to four drugs and the lowest quantity of strains being resistant to seven to eight drugs, with no significant differences being found among the four regions ($p > 0.05$). However, there was a significant difference in the multidrug resistance rate over two seasons, and the multidrug resistance rates with the three-to-four and seven-to-eight resistance patterns in spring were significantly higher than those same patterns in autumn ($p < 0.05$).

Antimicrobial resistance genotypes of the *S. aureus* isolates

The results for the antimicrobial resistance genes are presented in Fig. 3. In this study, 25 resistance genes were detected and were carried by 175 (63.4%) *S. aureus* isolates. The penicillin G-resistant gene (*blaZ*) was the gene with the highest detection rate at 46.3%, followed by the *dfcG* trimethoprim-resistance gene (35.5%) and the *tetM* tetracycline-resistance gene (27.2%). The remaining 22 resistance genes had detection rates that ranged from 3.3% to 23.6%. In addition, the *cfr* multidrug resistance gene was detected in 39 (14.1%) strains.

In our study, the majority of the *S. aureus* isolates possessed the *blaZ* gene in northeastern, northern, and southern China, where the detection rates were 47.8%, 45.3%, and 47.8%, respectively. The most prevalent gene detected in western China was *dfcG* (45.5%) (Table 5). The average detection rate for all 25 antibiotic resistance genes carried

TABLE 2. OCCURRENCE OF *STAPHYLOCOCCUS AUREUS* ISOLATED FROM RAW MILK SAMPLES IN THE DAIRY FARMS AMONG THE FOUR REGIONS BETWEEN SPRING AND AUTUMN IN CHINA

Region	No. of samples of all year (spring, autumn)	No. (%) of samples with confirmed <i>Staphylococcus aureus</i>		
		Spring	Autumn	All year
Northeastern China	200 (100, 100)	40 (40.0)	50 (50.0)	90 (45.0)
Northern China	240 (126, 114)	54 (42.9)	32 (28.1)	86 (35.8)
Western China	100 (30, 70)	13 (43.3)	20 (28.6)	33 (33.0)
Southern China	210 (109, 101)	43 (39.4)	24 (23.8)	67 (31.9)
Total	750 (365, 385)	150 (41.1)	126 (32.7)	276 (36.8)

TABLE 3. THE DISTRIBUTIONS OF ANTIMICROBIALS RESISTANCES IN PHENOTYPES OF *STAPHYLOCOCCUS AUREUS* ISOLATED FROM RAW MILK SAMPLES (N=276) IN THE DAIRY FARMS AMONG THE FOUR REGIONS BETWEEN SPRING AND AUTUMN IN CHINA

Antimicrobials	No. (%) of resistant isolates in different regions				No. (%) of resistant isolates in different seasons		Total (%)
	Northeastern China (n=90)	Northern China (n=86)	Western China (n=33)	Southern China (n=67)	Spring (n=150)	Autumn (n=126)	
β-Lactams							
Penicillin G	74 (82.2)	67 (77.9)	28 (84.8)	56 (83.6)	140 (93.3)	85 (67.5)	81.5
Oxacillin	60 (66.7)	52 (60.5)	13 (39.4)	49 (73.1)	105 (70.0)	69 (54.8)	63.0
Cefoxitin	29 (32.2)	29 (33.7)	7 (21.2)	11 (16.4)	40 (26.7)	36 (28.6)	27.5
Macrolides							
Erythromycin	31 (34.4)	34 (39.5)	8 (24.2)	29 (43.3)	73 (48.7)	29 (23.0)	37.0
Quinolones							
Ciprofloxacin	10 (11.1)	22 (25.6)	0 (0)	7 (10.4)	25 (16.7)	14 (11.1)	14.1
Tetracyclines							
Doxycycline	22 (24.4)	22 (25.6)	2 (6.1)	10 (14.9)	32 (21.3)	24 (19.0)	20.3
Tetracycline	21 (23.3)	28 (32.6)	4 (12.1)	10 (14.9)	38 (25.3)	25 (19.8)	22.8
Sulfonamides							
Sulfamethoxazole	59 (65.6)	69 (80.2)	23 (69.7)	52 (77.6)	133 (88.7)	70 (55.6)	73.6
Trimethoprim	60 (66.7)	67 (77.9)	20 (60.6)	47 (70.1)	126 (84.0)	68 (54.0)	70.3
Aminoglycosides							
Gentamicin	27 (30.0)	27 (31.4)	10 (30.3)	15 (22.4)	37 (24.7)	42 (33.3)	28.6
Kanamycin	29 (32.2)	32 (37.2)	13 (39.4)	15 (22.4)	49 (32.7)	40 (31.7)	32.2
Amikacin	36 (40.0)	29 (33.7)	9 (27.3)	5 (7.5)	29 (19.3)	50 (39.7)	28.6
Tobramycin	25 (27.8)	19 (22.1)	8 (24.2)	8 (11.9)	30 (20.0)	30 (23.8)	21.7
Lincosamides							
Clindamycin	51 (56.7)	43 (50)	18 (54.5)	27 (40.3)	83 (55.3)	56 (44.4)	50.4
Amphenicols							
Chloramphenicol	17 (18.9)	21 (24.4)	11 (33.3)	9 (13.4)	35 (23.3)	23 (18.3)	21.0
Average drug resistant rate (%)	40.8	43.5	35.1	34.8	43.3	35.0	39.5

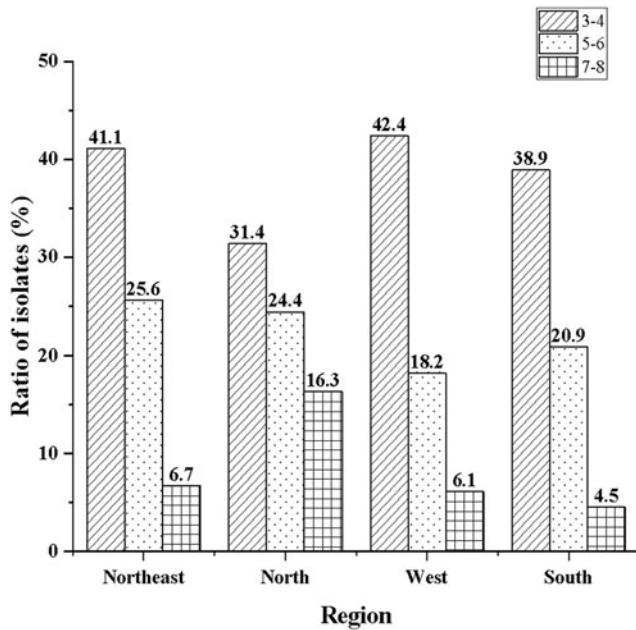


FIG. 1. The distributions of multiresistant to antimicrobials of *Staphylococcus aureus* isolated from raw milk samples in the dairy farms among the four regions of China. “3–4” means that the isolates resist to three to four classes of drugs; “5–6” means that the isolates resist to five to six classes of drugs; “7–8” means that the isolates resist to seven to eight classes of drugs.

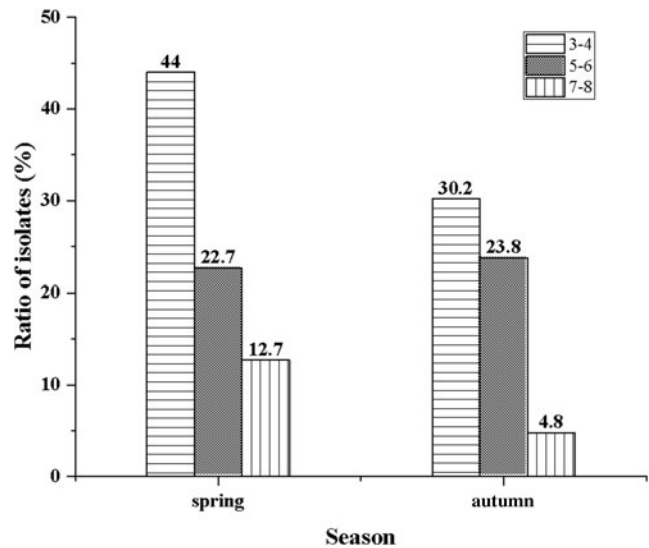


FIG. 2. The distributions of multiresistant to antimicrobials of *Staphylococcus aureus* isolated from raw milk samples in the dairy farms between spring and autumn. “3–4” means that the isolates resist to three to four classes of drugs; “5–6” means that the isolates resist to five to six classes of drugs; “7–8” means that the isolates resist to seven to eight classes of drugs.

TABLE 4. THE MAIN DRUG RESISTANCE PATTERNS OF 276 *STAPHYLOCOCCUS AUREUS* STRAINS ISOLATED FROM RAW MILK SAMPLES IN THE DAIRY FARMS AMONG THE FOUR REGIONS BETWEEN SPRING AND AUTUMN IN CHINA

No. of classes of drugs	Patterns of drug resistance	No. (%) of isolates in this pattern
0	—	15 (5.4)
1	βLs	19 (6.9)
	SULs	7 (2.5)
	MCs	1 (0.4)
	TCs	1 (0.4)
2	βLs-SULs	28 (10.1)
	βLs-QNs	3 (1.1)
	βLs-MCs	2 (0.7)
	βLs-LINs	2 (0.7)
	βLs-AGs	2 (0.7)
	SULs-AGs	2 (0.7)
	βLs-AMPs	1 (0.4)
3	βLs-SULs-AGs	12 (4.3)
	βLs-SULs-LINs	12 (4.3)
	βLs-MCs-SULs	9 (3.3)
	βLs-TCs-SULs	4 (1.4)
	βLs-SULs-AMPs	4 (1.4)
	βLs-MCs-LINs	3 (1.1)
	βLs-QNs-SULs	3 (1.1)
	SULs-AGs-LINs	2 (0.7)
	AGs-LINs-AMPs	2 (0.7)
	MCs-SULs-LINs	1 (0.4)
	QNs-SULs-AGs	1 (0.4)
	QNs-SULs-LINs	1 (0.4)
4	βLs-SULs-AGs-LINs	17 (6.2)
	βLs-MCs-SULs-LINs	8 (2.9)
	βLs-SULs-LINs-AMPs	5 (1.8)
	βLs-MCs-SULs-AGs	4 (1.4)
	βLs-MCs-QNs-LINs	3 (1.1)
	βLs-TCs-SULs-AGs	3 (1.1)
	βLs-TCs-SULs-LINs	3 (1.1)
	βLs-MCs-TCs-SULs	2 (0.7)
	βLs-QNs-SULs-AGs	1 (0.4)
	βLs-QNs-SULs-LINs	1 (0.4)
	βLs-TCs-SULs-AMPs	1 (0.4)
	βLs-SULs-AGs-AMPs	1 (0.4)
	SULs-AGs-LINs-AMPs	1 (0.4)
5	βLs-MCs-SULs-AGs-LINs	11 (4.0)
	βLs-SULs-AGs-LINs-AMPs	7 (2.5)
	βLs-TCs-SULs-AGs-LINs	6 (2.2)
	βLs-MCs-SULs-LINs-AMPs	3 (1.1)
	βLs-TCs-SULs-AGs-AMPs	3 (1.1)
	βLs-MCs-QNs-SULs-LINs	2 (0.7)
	βLs-MCs-TCs-SULs-LINs	2 (0.7)
	βLs-MCs-QNs-SULs-AGs	1 (0.4)
	βLs-MCs-TCs-SULs-AGs	1 (0.4)
6	βLs-MCs-TCs-SULs-AGs-LINs	9 (3.3)
	βLs-MCs-QNs-SULs-AGs-LINs	4 (1.4)
	βLs-MCs-SULs-AGs-LINs-AMPs	3 (1.1)
	βLs-TCs-SULs-AGs-LINs-AMPs	3 (1.1)
	βLs-MCs-QNs-TCs-SULs-AGs	2 (0.7)
	βLs-MCs-QNs-TCs-SULs-LINs	2 (0.7)
	βLs-MCs-TCs-SULs-AGs-AMPs	2 (0.7)
	βLs-MCs-QNs-SULs-AGs-AMPs	1 (0.4)
	βLs-MCs-TCs-SULs-LINs-AMPs	1 (0.4)
	βLs-QNs-TCs-SULs-AGs-LINs	1 (0.4)

(continued)

TABLE 4. (CONTINUED)

No. of classes of drugs	Patterns of drug resistance	No. (%) of isolates in this pattern
7	βLs-MCs-TCs-SULs-AGs-LINs-AMPs	13 (4.7)
	βLs-MCs-QNs-TCs-SULs-AGs-LINs	5 (1.8)
	βLs-MCs-QNs-TCs-SULs-AGs-AMPs	1 (0.4)
8	βLs-MCs-QNs-TCs-SULs-AGs-LINs-AMPs	6 (2.2)

The abbreviation of eight classes of drugs are as follows: βLs, β-lactams; MCs, macrolides; QNs, quinolones; TCs, tetracyclines; SULs, sulfonamides; AGs, aminoglycosides; LINs, lincosamides; AMPs, amphenicols. “—” means none.

by the *S. aureus* isolates was slightly higher in northern China (18.1%) than it was in southern (16.1%), northeastern (15.9%), and western China (15%), and no significant differences were found among the four regions ($p > 0.05$). For the two seasons, the incidence rate of the *blaZ* resistance gene in *S. aureus* isolates was higher in spring (52.7%) compared to autumn (38.9%). Similar to the resistant phenotype of *S. aureus*, the detection rate for all the resistant genes was higher in spring than in autumn, except for the aminoglycoside-resistant dominant genes of *ant* (6)-Ia and *ant* (4')-Ia. The average detection rate for the resistance genes was slightly higher in spring (18.8%) than in autumn (13.8%), and there was no significant difference between the two seasons ($p > 0.05$).

In our analysis, the average correlation coefficient between the resistant phenotype and genes of *S. aureus* was 64.8% (Table 6). The highest correlation coefficient was 84.5% for amphenicols, followed by 74.5% for macrolides, 74.0% for tetracyclines, 57.9% for β-lactams, 57.7% for sulfonamides, 52.8% for aminoglycosides, and 52.5% for lincosamides. The correlation between the resistant phenotypes and genes of *S. aureus* in the different regions are shown in Table 7. The correlation coefficient of the resistant phenotype and genes of *S. aureus* was the highest in northeastern China (66.5%), whereas it was the lowest in southern China (57.5%). In addition, the average correlation coefficient of the resistant phenotypes and genes of *S. aureus* were similar between the two seasons (Table 6).

Discussion

S. aureus is one of the main global pathogens involved in bovine mastitis in the dairy farms; thus, the presence of *S. aureus* in raw milk has received much attention.^{31,32} To the best of our knowledge, this study represents the first comprehensive report on the occurrence rates, drug-resistant phenotypes, and genotypes of *S. aureus* isolates in 750 raw milk samples obtained from 405 dairy farms in four regions across two seasons in China. The occurrence rates and characteristics of *S. aureus* isolated from raw milk in the dairy farms will be helpful to develop prevention and control procedures for pathogens specific to each region and season. In the present study, 36.8% (276/750) of the raw milk samples were positive for *S. aureus* of the 750 dairy

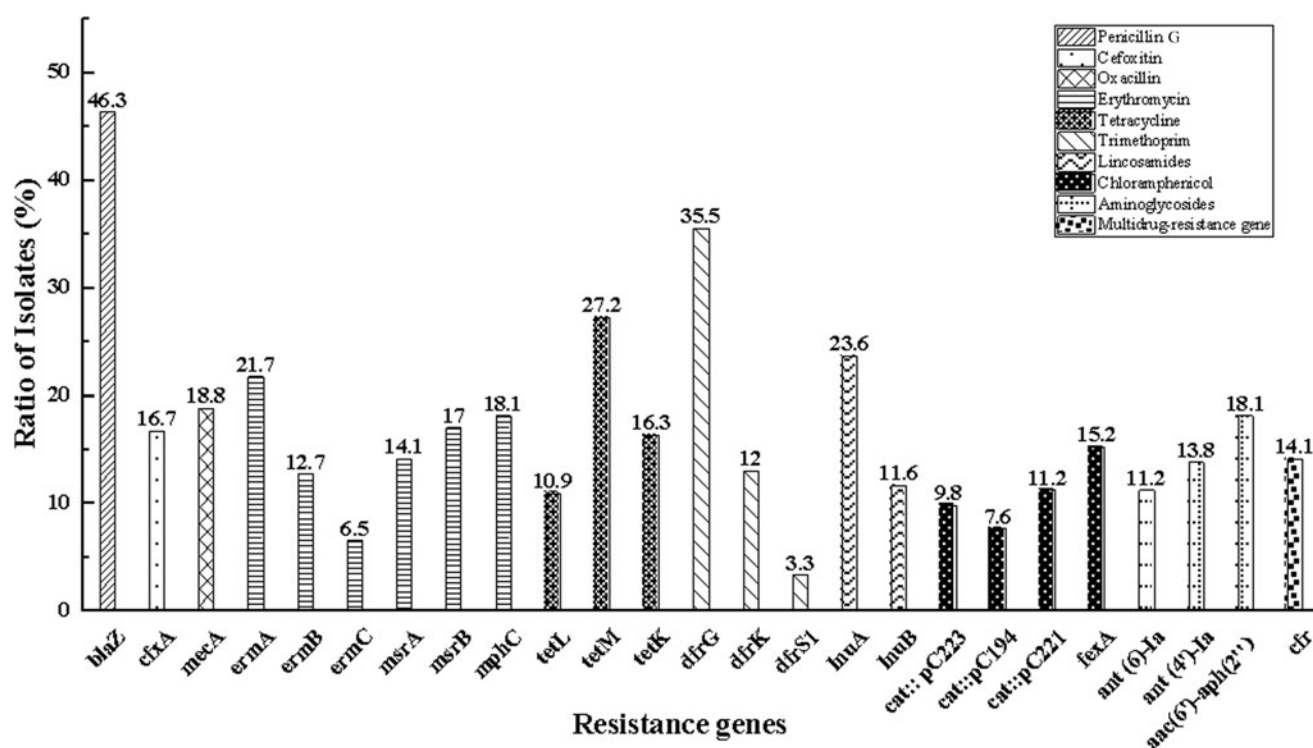


FIG. 3. The distributions of antibiotic resistance genes of *Staphylococcus aureus* isolated from raw milk samples.

samples collected from 405 dairy farms in China in 2016. This result was somewhat lower than those found in previous reports, which reported the occurrence rate of *S. aureus* in raw milk to be 83% in Italy,³³ 56% in Turkey,³⁴ and 47.3% in Norway.³⁵ However, our data are consistent with previous studies conducted in China, where the occurrence rate of *S. aureus* was 10.2–46.2% in raw mastitic milk samples.^{36–38} The isolation rate may be affected by the number of animals observed, stages of lactation, parity number, management practices, or other factors that are difficult to examine thoroughly.³⁹

Our records showed that the *S. aureus* isolation rate of the dairy farms was higher in spring than in autumn. As spring is a rainy season and autumn is a dry season in China,⁴⁰ elevated air temperatures and higher humidity in spring enhance bacterial growth, which may explain these results.⁴¹ Further studies need to be carried out to determine the exact cause of these differences. In our study, the isolation rate for *S. aureus* was higher in northeastern China than it was in northern, western, and southern China. Another report described similar observations regarding the *S. aureus* detection rate, which was found to be 50.1% in 894 bulk tank milk samples in northeastern China.²⁰ Although the climate in northeastern China is mostly cool and dry, more *S. aureus* isolates were found in that region.

Several previous studies have indicated that the rate of dairy cow infections with pathogens was related to geographic variations as well as the management of the herds under investigation (e.g., periparturient dairy cows and calving management, bedding maintenance, and cleaning rates for the cows, etc.).^{38,42} In addition, healthy cows and infected cows are often milked together using the same milking machinery on several dairy-farming communities in

northeastern China, which may explain our findings. Thus, farm management practices could be introduced to protect dairy cows against contagious mastitis pathogens, which would complement necessary antimicrobial-resistance procedures to determine the most appropriate veterinary drugs for mastitis treatment.

In the present study, 94.6% of *S. aureus* isolates from raw milk obtained from dairy farms showed resistance to at least one antibiotic, and 69.9% of the strains showed multidrug resistance (resistant to at least one agent in at least three antimicrobial classes). Our results for drug resistance were much higher than those described in other reports. For example, a previous study reported that 9.09% of the *S. aureus* isolates in milk were defined as multidrug resistant in China.¹⁸ Another study found that 15.4% of the *S. aureus* isolates in raw milk in Iran were multidrug resistant.¹⁷ The high occurrence of multidrug resistance in *S. aureus* from raw milk is a potential risk to human health.

Most *S. aureus* isolates were found to have high levels of resistance to penicillin G, sulfamethoxazole, trimethoprim, oxacillin, and clindamycin; however, there was a much lower level of resistance to ciprofloxacin. In particular, the resistance rate of penicillin G was similar between north-west China (84.1%)¹⁸ and northern China (85.2%).¹⁹ Prior studies in other countries have also shown that there is a high occurrence of penicillin G-resistant *S. aureus* isolates in raw milk.^{17,22} The high resistance rate of penicillin G-resistant *S. aureus* isolates may likely be related to the extensive use of antibiotics to treat infections in the dairy farms. Recognizing the resistance characteristics of *S. aureus* is essential for monitoring the spread of resistant strains and controlling the potential harm linked to pathogens.⁴³ The multidrug resistance patterns of *S. aureus* in

TABLE 5. THE DISTRIBUTIONS OF ANTIBIOTIC RESISTANCE GENES OF *STAPHYLOCOCCUS AUREUS* ISOLATED FROM RAW MILK SAMPLES IN THE DAIRY FARMS AMONG THE FOUR REGIONS BETWEEN SPRING AND AUTUMN IN CHINA

		No. (%) of resistant isolates in different regions				No. (%) of resistant isolates in different seasons		Total (%)
		Northeastern China (n=90)	Northern China (n=86)	Western China (n=33)	Southern China (n=67)	Spring (n=150)	Autumn (n=126)	
<i>Antimicrobials</i>								
Penicillin G	<i>blaZ</i>	43 (47.8)	39 (45.3)	14 (42.4)	32 (47.8)	79 (52.7)	49 (38.9)	128 (46.3)
Cefoxitin	<i>cfxA</i>	14 (15.6)	18 (20.9)	3 (9.1)	11 (16.4)	32 (21.3)	14 (11.1)	46 (16.7)
Oxacillin	<i>mecA</i>	17 (18.9)	21 (24.4)	2 (6.1)	12 (17.9)	33 (22.0)	19 (15.1)	52 (18.8)
Erythromycin	<i>ermA</i>	21 (23.3)	22 (25.6)	4 (12.1)	13 (19.4)	44 (29.3)	16 (12.7)	60 (21.7)
	<i>ermB</i>	10 (11.1)	12 (14.0)	4 (12.1)	9 (13.4)	21 (14.0)	14 (11.1)	35 (12.7)
	<i>ermC</i>	6 (6.7)	4 (4.7)	1 (3.0)	7 (10.4)	11 (7.3)	7 (5.6)	18 (6.5)
	<i>msrA</i>	13 (14.4)	16 (18.6)	3 (9.1)	7 (10.4)	25 (16.7)	14 (11.1)	39 (14.1)
Tetracycline	<i>msrB</i>	19 (21.1)	20 (23.3)	5 (15.2)	3 (4.5)	31 (20.7)	16 (12.7)	47 (17.0)
	<i>mphC</i>	18 (20.0)	13 (15.1)	2 (6.1)	17 (25.3)	30 (20.0)	20 (15.9)	50 (18.1)
	<i>tetL</i>	8 (8.9)	14 (16.3)	3 (9.1)	5 (7.5)	21 (14.0)	9 (7.1)	30 (10.9)
	<i>tetM</i>	21 (23.3)	25 (29.1)	6 (18.2)	23 (34.3)	45 (30.0)	30 (23.8)	75 (27.2)
Trimethoprim	<i>tetK</i>	7 (7.8)	18 (20.9)	8 (24.2)	12 (17.9)	28 (18.7)	17 (13.5)	45 (16.3)
	<i>dfrG</i>	28 (31.1)	35 (40.7)	15 (45.5)	20 (29.9)	54 (36.0)	44 (34.9)	98 (35.5)
	<i>dfrK</i>	11 (12.2)	9 (10.5)	5 (15.2)	8 (11.9)	20 (13.3)	13 (10.3)	33 (12.0)
	<i>dfrS1</i>	4 (4.4)	3 (3.5)	0 (0)	2 (3.0)	5 (3.3)	4 (3.2)	9 (3.3)
Lincosamide	<i>lnuA</i>	22 (24.4)	27 (31.4)	7 (21.2)	9 (13.4)	42 (28.0)	23 (18.3)	65 (23.6)
	<i>lnuB</i>	10 (11.1)	15 (17.4)	3 (9.1)	4 (6.0)	24 (16.0)	8 (6.3)	32 (11.6)
Chloramphenicol	<i>cat::pC223</i>	8 (8.9)	10 (11.6)	4 (12.1)	5 (7.5)	17 (11.3)	10 (7.9)	27 (9.8)
	<i>cat::pC194</i>	7 (7.8)	5 (5.8)	1 (3.0)	8 (11.9)	15 (10.0)	6 (4.8)	21 (7.6)
	<i>cat::pC221</i>	10 (11.1)	7 (8.1)	5 (15.2)	9 (13.4)	21 (14.0)	10 (7.9)	31 (11.2)
	<i>fexA</i>	13 (14.4)	12 (14.0)	5 (15.2)	12 (17.9)	25 (16.7)	17 (13.5)	42 (15.2)
Aminoglycosides	<i>ant (6)-Ia</i>	9 (10.0)	9 (10.5)	3 (9.1)	10 (14.9)	10 (6.7)	21 (16.7)	31 (11.2)
	<i>ant (4')-Ia</i>	14 (15.6)	11 (12.8)	7 (21.2)	6 (9.0)	18 (12.0)	20 (15.9)	38 (13.8)
	<i>aac(6')-aph(2'')</i>	14 (15.6)	15 (17.4)	6 (18.2)	15 (22.4)	28 (18.7)	22 (17.5)	50 (18.1)
Multidrug resistance gene	<i>cfr</i>	11 (12.2)	10 (11.6)	8 (24.2)	10 (14.9)	27 (18.0)	12 (9.5)	39 (14.1)
Average gene detection rate (%)		15.9	18.1	15.0	16.1	18.8	13.8	16.5

the four regions of China were basically similar, with the highest number of strains being resistant to three to four classes of drugs, and the multidrug resistance rate was higher in spring than in autumn. This is the first article to investigate multidrug resistance by *S. aureus* across different seasons and regions in China. This information will guide the adoption of appropriate antibiotic treatments (such as ciprofloxacin) for mastitis caused by *S. aureus* in different seasons and regions in China.

Just as there was a higher isolation rate for *S. aureus* in spring than there was in autumn, the resistance rate of *S. aureus* to most veterinary drugs also showed the same trend. In a previous study, *S. aureus* isolates in milk were significantly associated with the season, and the same association was observed for resistant strains.⁴⁴ This phenomenon could be related to the repeated therapeutic use of veterinary drugs. At the same time, the veterinary drugs used to treat mastitis in dairy cows were similar among the dairy farms throughout the year, according to veterinary records. In addition, *S. aureus* isolated from dairy farms in the four regions of China showed similar average resistances to eight classes of antimicrobials tested in this study. The average drug resistance rate of *S. aureus* isolated from raw

milk samples was slightly lower in southern China than it was in the other three regions. Our study revealed that the incidence rates of *S. aureus* in raw milk were different among the four regions. This phenomenon may be related to the differences in agricultural practices and also climatic conditions between these regions.⁴⁵ As discussed previously, the distribution and variation in resistance genes of *S. aureus* strains isolated from raw milk from dairy farms were first mapped in major regions over two seasons in China. As a result, this study may contribute to the development of appropriate strategies to prevent dairy cows from being infected in different areas.

In our study, 25 different resistance genes were identified in the 175 *S. aureus* isolates. The most prevalent resistance genes were *blaZ* (46.3%), *dfrG* (35.5%), and *tetM* (27.2%). A recent study found that the *blaZ*, *ermB*, *rpoB*, *ermC*, *tetK*, and *tetM* resistance genes in the *S. aureus* isolates were highly prevalent in northwest China.¹⁸ Furthermore, a study also reported that the resistance genes of *aac(6')-aph(2'')*, *ant(4')-Ia*, *mecA*, *blaZ*, and *tetM* genes from *S. aureus* strains have high occurrence rates in northern China.¹⁹ In Brazil, the most prevalent resistance genes in *S. aureus* isolates were *tetK*, *tetL*, and *ereB*.⁴⁶ In Iran, the *blaZ*, *tetM*,

TABLE 6. THE CORRELATION BETWEEN RESISTANT PHENOTYPES AND GENES OF *STAPHYLOCOCCUS AUREUS* ISOLATED FROM RAW MILK SAMPLES IN THE DAIRY FARMS BETWEEN SPRING AND AUTUMN

Antimicrobial classes	No. of resistant phenotypes and genes of <i>Staphylococcus aureus</i>									
	Spring					Autumn				
	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)	Total no. of resistance phenotypes	Total no. of resistance genes	Total correlation coefficients (%)	
β -Lactams	144	86	59.7	98	54	55.1	242	140	57.9	
Macrolides	73	54	74.0	29	22	75.9	102	76	74.5	
Tetracyclines	43	57	75.4	28	39	71.8	71	96	74.0	
Sulfonamides	145	78	53.8	77	50	64.9	222	128	57.7	
Aminoglycosides	64	39	60.9	63	28	44.4	127	67	52.8	
Lincosamides	83	42	50.6	56	31	55.4	139	73	52.5	
Amphenicols	35	29	82.9	23	20	87.0	58	49	84.5	
Average			65.3			64.9			64.8	

TABLE 7. THE CORRELATION BETWEEN RESISTANT PHENOTYPES AND GENES OF *STAPHYLOCOCCUS AUREUS* ISOLATED FROM RAW MILK SAMPLES IN THE DAIRY FARMS AMONG THE FOUR REGIONS OF CHINA

Antimicrobial classes	No. of resistant phenotypes and genes of <i>Staphylococcus aureus</i>											
	Northeastern China				Northern China				Western China			
	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)	No. of resistance phenotypes	No. of resistance genes	Correlation coefficients (%)
β -Lactams	77	45	58.4	79	48	60.8	28	15	53.6	58	32	55.2
Macrolides	31	25	80.6	34	23	67.6	8	5	62.5	29	23	79.3
Tetracyclines	25	22	88.0	29	27	93.1	5	10	50.0	12	37	32.4
Sulfonamides	67	36	53.7	75	46	61.3	26	17	65.4	54	29	53.7
Aminoglycosides	43	19	44.2	47	21	44.7	17	10	58.8	20	17	85.0
Lincosamides	51	27	53.0	43	27	62.8	18	9	50.0	27	10	37.0
Amphenicols	17	15	88.2	21	12	57.1	11	7	63.6	9	15	60.0
Average			66.5			63.9			57.7			57.5

and *ermC* resistance genes in *S. aureus* isolates in mastitis milk were found at rates of 86%, 39.5%, and 39.5%, respectively.⁴⁷ Other articles from Iran have indicated that high frequencies of *blaZ* (46%) and *tetM* (34.8%) resistance genes were present in *S. aureus* isolates.¹⁷ These findings indicate that the most prevalent resistance gene was *blaZ*, which was in agreement with earlier findings. However, the occurrence rate of the *dfrG* was different from earlier findings, which may be caused by the different medication habits on different dairy farms.⁴³ In the current study, our results showed the average resistance gene detection rate of *S. aureus* strains in northern China was slightly higher than those for all other regions. We assume that this difference might be caused by cows that have not responded to antibiotic treatment, which has not completely eliminated the causative pathogens.

In the present study, our results showed that the resistant phenotypes and genotypes of *S. aureus* were not completely consistent. The high correlation coefficient was 84.5% for amphenicols, 74.5% for macrolides, and 74.0% for tetracyclines. These data were similar to those in another report that identified a moderate correlation between the resistant phenotype and genotype for tetracyclines.¹⁸ However, a different study found that none of the tetracycline-resistant *S. aureus* strains carried the *tetM* gene⁴⁸; one explanation for this finding is that phenotypic resistance may not be caused by genetic acquisition, or the “intrinsic” mechanisms of bacterial resistance (such as biofilms or multidrug efflux pumps) may play a primary role in resistance.^{18,49} Future research should continue to investigate the relationship between resistance genes and phenotypes and develop new strategies to combat antimicrobial resistance, including the prevention and treatment of bovine mastitis by examining phenotypic susceptibility to specific antimicrobial resistance genes.

Conclusions

In this study, we investigated *S. aureus* in raw milk obtained from dairy farms in four different regions over two seasons in China. Our data revealed that the detected *S. aureus* isolates had a high percentage of antimicrobial resistance and carried multiple antimicrobial resistance genes. Therefore, it is of considerable significance to develop treatment methods based on the different regional and seasonal features relating to the incidence and resistance of *S. aureus* strains. The findings will help prevent infections with *S. aureus* in cows on dairy farms and further protect the health of the end consumers.

Disclosure Statement

No competing financial interests exist.

Funding Information

The authors gratefully acknowledge the Key R&D Program of Shandong Province (2019GNC106024), Scientific Research Foundation of the Higher Education Institutions of Shandong Province (J17KA131), Breeding Plan of Shandong Provincial Qingchuang Research Team (2019), the Special Fund for Agro-scientific Research in the Public Interest (201403071), National risk assessment major special

project of milk product quality and safety (GJFP2019027), Agricultural Technology Experimental Demonstration and Service Support Project in 2018 (15182130106407160), and Natural Science Foundation of Shandong Province of the P.R. China (ZR2015CQ020).

References

1. Conceição, T., H. de Lencastre, and M. Aires-de-Sousa. 2017. Healthy bovines as reservoirs of major pathogenic lineages of *Staphylococcus aureus* in Portugal. *Microb. Drug Resist.* 23:845–851.
2. Eisenberg, S.W.F., E.M. Boerhout, L. Ravesloot, A.J.J.M. Daemen, L. Benedictus, V.P.M.G. Rutten, and A.P. Koets. 2016. Diurnal differences in milk composition and its influence on in vitro growth of *Staphylococcus aureus* and *Escherichia coli* in bovine quarter milk. *J. Dairy Sci.* 99: 5690–5700.
3. Da Costa Krewer, C., E. Santos Amanso, G. Veneroni Gouveia, R. de Lima Souza, M.M. da Costa, and R. Aparecido Mota. 2015. Resistance to antimicrobials and biofilm formation in *Staphylococcus* spp. isolated from bovine mastitis in the Northeast of Brazil. *Trop. Anim. Health Prod.* 47:511–518.
4. Kateete, D.P., U. Kabugo, H. Baluku, L. Nyakarahuka, S. Kyobe, M. Okee, C.F. Najjuka, and M.L. Joloba. 2013. Prevalence and antimicrobial susceptibility patterns of bacteria from milkmen and cows with clinical mastitis in and around Kampala, Uganda. *PLoS One* 8:e63413.
5. Feßler, A.T., H. Kaspar, C.J. Lindeman, M.R. Stegemann, T. Peters, J. Mankertz, J.L. Watts, and S. Schwarz. 2012. A proposal of interpretive criteria for cefoperazone applicable to bovine mastitis pathogens. *Vet. Microbiol.* 157: 226–231.
6. Kumar, R., B.R. Yadav, and R.S. Singh. 2011. Antibiotic resistance and pathogenicity factors in *Staphylococcus aureus* isolated from mastitic Sahiwal cattle. *J. Biosci.* 36: 175–188.
7. Haran, K.P., S.M. Godden, D. Boxrud, S. Jawahir, J.B. Bender, and S. Sreevatsan. 2012. Prevalence and characterization of *Staphylococcus aureus*, including methicillin-resistant *Staphylococcus aureus*, isolated from bulk tank milk from Minnesota dairy farms. *J. Clin. Microbiol.* 50: 688–695.
8. Caggiano, G., A. Dambrosio, F. Ioanna, S. Balbino, G. Barbuti, O. De Giglio, G. Diella, G. Lovero, S. Rutigliano, G. Scarafile, A. Baldassarre, L. Vimercati, M. Musti, and M.T. Montagna. 2016. Prevalence and characterization of methicillin-resistant *Staphylococcus aureus* isolates in food industry workers. *Ann. Ig.* 28:8–14.
9. Xing, X., Y. Zhang, Q. Wu, X. Wang, W. Ge, and C. Wu. 2016. Prevalence and characterization of *Staphylococcus aureus* isolated from goat milk powder processing plants. *Food Control* 59:644–650.
10. Tarekgne, E., S. Skeie, K. Rudi, T. Skjerdal, and J.A. Narvhus. 2015. *Staphylococcus aureus* and other *Staphylococcus* species in milk and milk products from Tigray region, Northern Ethiopia. *Afr. J. Food Sci.* 9:567–576.
11. Bengtsson, B., H.E. Unnerstad, T. Ekman, K. Artursson, M. Nilsson-Ost, and K.P. Waller. 2009. Antimicrobial susceptibility of udder pathogens from cases of acute clinical mastitis in dairy cows. *Vet. Microbiol.* 136:142–149.
12. Ruegg, P.L., L. Oliveira, W. Jin, and O. Okwumabua. 2015. Phenotypic antimicrobial susceptibility and occurrence of

- selected resistance genes in gram-positive mastitis pathogens isolated from Wisconsin dairy cows. *J. Dairy Sci.* 98: 4521–4534.
13. Saini, V., J.T. McClure, D. Léger, S. Dufour, A.G. Sheldon, D.T. Scholl, and H.W. Barkema. 2012. Antimicrobial use on Canadian dairy farms. *J. Dairy Sci.* 95:1209–1221.
 14. Szweda, P., M. Schielmann, A. Frankowska, B. Kot, and M. Zalewska. 2014. Antibiotic resistance in *Staphylococcus aureus* strains isolated from cows with mastitis in eastern Poland and analysis of susceptibility of resistant strains to alternative nonantibiotic agents: lysostaphin, nisin and polymyxin B. *J. Vet. Med. Sci.* 76:355–362.
 15. Li, J., H. Zhou, L. Yuan, T. He, and S. Hu. 2009. Prevalence, genetic diversity, and antimicrobial susceptibility profiles of *Staphylococcus aureus* isolated from bovine mastitis in Zhejiang Province, China. *J. Zhejiang Univ. Sci. B.* 10:753–760.
 16. Wendlandt, S., A.T. Feßler, S. Monecke, R. Ehricht, S. Schwarz, and K. Kadlec. 2013. The diversity of antimicrobial resistance genes among staphylococci of animal origin. *Int. J. Med. Microbiol.* 303:338–349.
 17. Jamali, H., M. Paydar, B. Radmehr, and S. Ismail. 2015. Prevalence and antimicrobial resistance of *Staphylococcus aureus* isolated from raw milk and dairy products. *Food Control* 54:383–388.
 18. Yang, F., Q. Wang, X. Wang, L. Wang, X. Li, J. Luo, S. Zhang, and H. Li. 2016. Genetic characterization of antimicrobial resistance in *Staphylococcus aureus* isolated from bovine mastitis cases in Northwest China. *J. Integr. Agr.* 15:2842–2847.
 19. Liu, H., S. Li, L. Meng, L. Dong, S. Zhao, X. Lan, J. Wang, and N. Zheng. 2017. Prevalence, antimicrobial susceptibility, and molecular characterization of *Staphylococcus aureus* isolated from dairy herds in Northern China. *J. Dairy Sci.* 100:8796–8803.
 20. Bi, Y., Y.J. Wang, Q. Yun, R. Guix Vallverdú, J. Maldonado García, W. Sun, S. Li, and Z. Cao. 2016. Prevalence of bovine mastitis pathogens in bulk tank milk in China. *PLoS One.* 11:e0155621.
 21. Clinical and Laboratory Standards Institute. 2014. Performance standards for antimicrobial susceptibility testing; 24th edition. CLSI document M100-S24. CLSI, Wayne, PA.
 22. Zehra, A., R. Singh, S. Kaur, and J.P.S. Gill. 2017. Molecular characterization of antibiotic-resistant *Staphylococcus aureus* from livestock (bovine and swine). *Vet. World* 10: 598–604.
 23. Avelar, K.E.S., K. Otsuki, A.C.P. Vicente, J.M.B.D. Vieira, G.R. de Paula, R.M.C.P. Domingues, and M.C.D.S. Ferreira. 2003. Presence of the *cfx A* gene in *Bacteroides distasonis*. *Res. Microbiol.* 154:369–374.
 24. Strommenger, B., C. Kettlitz, G. Werner, and W. Witte. 2003. Multiplex PCR assay for simultaneous detection of nine clinically relevant antibiotic resistance genes in *Staphylococcus aureus*. *J. Clin. Microbiol.* 41:4089–4094.
 25. Pekana, A., and E. Green. 2018. Antimicrobial resistance profiles of *Staphylococcus aureus* isolated from meat carcasses and bovine milk in abattoirs and dairy farms of the Eastern Cape, South Africa. *Int. J. Environ. Res. Public Health.* 15:pii:E2223.
 26. Argudín, M.A., B.-A. Tenhagen, A. Fetsch, J. Sachsenrder, A. Kasbohrer, A. Schroeter, J.A. Hammerl, S. Hertwig, R. Helmuth, J. Braunig, M.C. Mendoza, B. Appel, M. R. Rodicio, and B. Guerra. 2011. Virulence and resistance determinants of German *Staphylococcus aureus* ST398 isolates from nonhuman sources. *Appl. Environ. Microbiol.* 77:3052–3060.
 27. Lüthje, P., and S. Schwarz. 2006. Antimicrobial resistance of coagulase-negative staphylococci from bovine subclinical mastitis with particular reference to macrolide-lincosamide resistance phenotypes and genotypes. *J. Antimicrob. Chemother.* 57:966–969.
 28. Lina, G., A. Quaglia, M.E. Reverdy, R. Leclercq, F. Vandenesch, and J. Etienne. 1999. Distribution of genes encoding resistance to macrolides, lincosamides, and streptogramins among *Staphylococci*. *Antimicrob. Agents Chemother.* 43:1062.
 29. Bozdogan, B., L. Berrezouga, M.-S. Kuo, D.A. Yurek, K.A. Farley, B.J. Stockman, and R. Leclercq. 1999. A new resistance gene, *linB*, conferring resistance to lincosamides by nucleotidylation in *Enterococcus faecium* HM1025. *Antimicrob. Agents Chemother.* 43:925–929.
 30. Kehrenberg, C., and S. Schwarz. 2006. Distribution of florfenicol resistance genes *fexA* and *cfr* among chloramphenicol-resistant *Staphylococcus* isolates. *Antimicrob. Agents Chemother.* 50:1156–1163.
 31. Le Maréchal, C., N. Seyffert, J. Jardin, D. Hernandez, G. Jan, L. Rault, V. Azevedo, P. François, J. Schrenzel, M. van de Guchte, S. Even, N. Berkova, R. Thiery, J.R. Fitzgerald, E. Vautor, and Y. Le Loir. 2011. Molecular basis of virulence in *Staphylococcus aureus* mastitis. *PLoS One* 6:e27354.
 32. Riva, A., E. Borghi, D. Cirasola, S. Colmegna, F. Borgo, E. Amato, M.M. Pontello, and G. Morace. 2015. Methicillin-resistant *Staphylococcus aureus* in raw milk: prevalence, SCCmec typing, enterotoxin characterization, and antimicrobial resistance patterns. *J. Food Protect.* 78: 1142–1146.
 33. Bartolomeoli, I., M. Maifreni, F. Frigo, G. Urli, and M. Marino. 2009. Occurrence and characterization of *Staphylococcus aureus* isolated from raw milk for cheesemaking. *Int. J. Dairy Technol.* 62:366–371.
 34. Gundogan, N., and E. Avci. 2014. Occurrence and antibiotic resistance of *Escherichia coli*, *Staphylococcus aureus* and *Bacillus cereus* in raw milk and dairy products in Turkey. *Int. J. Dairy Technol.* 67:562–569.
 35. Jakobsen, R.A., R. Heggeb, E.B. Sunde, and M. Skjervheim. 2011. *Staphylococcus aureus* and *Listeria monocytogenes* in Norwegian raw milk cheese production. *Food Microbiol.* 28:492–496.
 36. Wang, W., X. Lin, T. Jiang, Z. Peng, J. Xu, L. Yi, F. Li, S. Fanning, and Z. Baloch. 2018. Prevalence and characterization of *Staphylococcus aureus* cultured from raw milk taken from dairy cows with mastitis in Beijing, China. *Front. Microbiol.* 9:1123.
 37. Wang, D., L. Zhang, X. Zhou, Y. He, C. Yong, M. Shen, O. Szenci, and B. Han. 2016. Antimicrobial susceptibility, virulence genes, and randomly amplified polymorphic DNA analysis of *Staphylococcus aureus* recovered from bovine mastitis in Ningxia, China. *J. Dairy Sci.* 99:9560–9569.
 38. Gao, J., H.W. Barkema, L. Zhang, G. Liu, Z. Deng, L. Cai, R. Shan, S. Zhang, J. Zou, J.P. Kastelic, and B. Han. 2017. Incidence of clinical mastitis and distribution of pathogens on large Chinese dairy farms. *J. Dairy Sci.* 100:4797–4806.
 39. Getaneh, A.M., and E.Z. Gebremedhin. 2017. Meta-analysis of the prevalence of mastitis and associated risk

- factors in dairy cattle in Ethiopia. *Trop. Anim. Health Prod.* 49:697–705.
40. Tong, L., S. Huang, Y. Wang, H. Liu, and M. Li. 2014. Occurrence of antibiotics in the aquatic environment of Jiangnan Plain, central China. *Sci. Total Environ.* 497–498: 180–187.
 41. Zhang, Z., X.P. Li, F. Yang, J.Y. Luo, X.R. Wang, L.H. Liu, and H.S. Li. 2016. Influences of season, parity, lactation, udder area, milk yield, and clinical symptoms on intramammary infection in dairy cows. *J. Dairy Sci.* 99: 6484–6493.
 42. Riekerink, R.G.M.O., H.W. Barkema, D.T. Scholl, D.E. Poole, and D.F. Kelton. 2010. Management practices associated with the bulk-milk prevalence of *Staphylococcus aureus* in Canadian dairy farms. *Prev. Vet. Med.* 97:20–28.
 43. Abdi, R.D., B.E. Gillespie, J. Vaughn, C. Merrill, S.I. Headrick, D.B. Ensermu, D.H. D'Souza, G.E. Agga, R.A. Almeida, S.P. Oliver, and O.K. Dego. 2018. Antimicrobial resistance of *Staphylococcus aureus* isolates from dairy cows and genetic diversity of resistant isolates. *Foodborne Pathog. Dis.* 15:449–458.
 44. Østeras, O., L. Sølverød, and O. Reksen. 2006. Milk culture results in a large norwegian survey—Effects of season, parity, days in milk, resistance, and clustering. *J. Dairy Sci.* 89:1010–1023.
 45. Jørgensen, H.J., T. Mørk, H.R. Høgåsen, and L.M. Rørvik. 2005. Enterotoxigenic *Staphylococcus aureus* in bulk milk in Norway. *J. Appl. Microbiol.* 99:158–166.
 46. Haubert, L., I.S. Kroning, M.A. Iglesias, and W.P. da Silva. 2017. First report of the *Staphylococcus aureus* isolate from subclinical bovine mastitis in the South of Brazil harboring resistance gene *dfrG* and transposon family Tn 916–1545. *Microb. Pathogenesis* 113:242–247.
 47. Jamali, H., B. Radmehr, and S. Ismail. 2014. Short communication: prevalence and antibiotic resistance of *Staphylococcus aureus* isolated from bovine clinical mastitis. *J. Dairy Sci.* 97:2226–2230.
 48. Shamila-Syuhada, A.K., G. Rusul, W.A. Wan-Nadiah, and L.-O. Chuah. 2016. Prevalence and antibiotics resistance of *Staphylococcus aureus* isolates isolated from raw milk obtained from small-scale dairy farms in Penang, Malaysia. *Pak. Vet. J.* 36:98–102.
 49. Lowy, F.D. 2003. Antimicrobial resistance: the example of *Staphylococcus aureus*. *J. Clin. Invest.* 111:1265–1273.

Address correspondence to:

Rongwei Han, PhD

College of Food Science and Engineering

Qingdao Agricultural University

No. 700 Changcheng Road

Qingdao City 265200

Shandong Province

China

E-mail: qauhan@qau.edu.cn